TECHNICAL REPORT

A Multimodal Approach for Alzheimer's Disease Detection
from MRI and Clinical Data

Using the OASIS-1 Dataset

Graduation Project | May 2026

1. Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder affecting over 55 million people worldwide. Early and accurate detection across its distinct clinical stages is critical for enabling timely intervention. Current deep learning approaches for AD classification predominantly rely on a single data modality, either neuroimaging or clinical assessments, which limits their diagnostic capacity.

This report presents a multimodal deep learning model that jointly processes 3D structural T1-weighted MRI volumes and clinical tabular data to classify subjects into three AD severity categories. The proposed architecture introduces a cross-modal attention fusion mechanism in which clinical features act as a query over spatial brain regions, enabling the model to learn which anatomical areas are most relevant given a patient's clinical profile.

The work is conducted on the OASIS-1 dataset and is directly inspired by the hybrid Transformer-CNN architecture of Alorf (2025), which demonstrated strong multimodal classification on rs-fMRI data. This project adapts and extends that design to structural MRI, a more accessible and widely available imaging modality.

2. Dataset

2.1 OASIS-1 Overview

The Open Access Series of Imaging Studies (OASIS-1) is a publicly available cross-sectional dataset comprising structural MRI scans and clinical assessments of 436 subjects ranging from young to older adults, including both cognitively normal individuals and those with varying degrees of dementia.

After matching NIfTI files with CSV records, 416 subjects were retained for this study. The dataset was split at the subject level (not the scan level) to prevent data leakage from the 20 subjects with repeat scans (MR2 suffix). A stratified 80/20 train-validation split yielded 332 training and 84 validation subjects.

Property	Value	Property	Value
Total subjects	436 (416 usable)	Training set	332 subjects
MRI modality	T1-weighted structural	Validation set	84 subjects
MRI space	MNI152 2mm (91x109x91)	Number of classes	3
Clinical features	7 (Age, Sex, Educ, SES, MMSE, eTIV, nWBV)	Label source	CDR score

Table 1. OASIS-1 dataset summary.

2.2 Label Definition

Subjects were classified based on the Clinical Dementia Rating (CDR) scale. Since only 5 subjects carried a CDR score of 2.0 (Moderate dementia), CDR 1.0 and 2.0 were merged to form a single Mild/Moderate class, yielding a three-class problem:

Class Index	Label	CDR Value	Train Subjects (n)
0	Non-demented	0.0	252 (75.9%)
1	Very mild dementia	0.5	56 (16.9%)
2	Mild / Moderate dementia	1.0 + 2.0	24 (7.2%)

Table 2. Class definitions and training set distribution.

The dataset is significantly imbalanced, with the Non-demented class comprising 75.9% of training samples. This was addressed by applying inverse-frequency class weights to the loss function during training.

3. Data Preprocessing

3.1 MRI Preprocessing

The NIfTI files in the dataset were already extensively preprocessed. The following steps had been applied prior to model training:

- DICOM to NIfTI conversion using dcm2niix and fslchfiletype
- Linear registration to MNI152 2mm template using FLIRT (12 degrees of freedom, trilinear interpolation)
- Tissue segmentation with FAST (3 tissue classes)
- N4 bias field correction using ANTs (version 2.2.0)
- Brain extraction using BET2 (fractional intensity 0.3)
- White matter intensity normalization: each volume divided by the mean WM voxel value

Within the training pipeline, the following additional steps were applied per volume:

- Canonical RAS+ reorientation using nibabel as_closest_canonical
- Resampling to a uniform 96x96x96 voxel grid using trilinear interpolation
- Intensity clipping at the 1st and 99th percentile, then rescaling to [0, 1]
- Addition of a channel dimension, producing shape (1, 96, 96, 96)

Data augmentation was applied to the training set only: random left-right flip (p=0.5), random rotation up to 10 degrees (p=0.5), and random intensity scaling of 5% (p=0.5). No augmentation was applied during validation or inference.

3.2 Clinical Feature Preprocessing

Seven clinical features were selected from the OASIS-1 CSV file. The following columns were excluded: ID (identifier), CDR (label), Hand (near-zero variance), ASF (collinear with eTIV), and Delay (missing in 95% of subjects).

The selected features are: Age, Sex (M/F encoded as 0/1), Education level (Educ), Socioeconomic status (SES), Mini-Mental State Examination score (MMSE), Estimated total intracranial volume (eTIV), and Normalized whole brain volume (nWBV).

Missing values in SES and Educ were imputed using median imputation. All features were standardized using StandardScaler fitted on the training set only, then applied to the validation set. Class weights for the loss function were computed as inverse class frequencies and applied to nn.CrossEntropyLoss.

4. Model Architecture

The proposed model, AlzheimerFusionModel, consists of four components: a 3D MRI encoder, a clinical MLP encoder, a cross-modal fusion module, and a classifier head. The overall design is inspired by the hybrid Transformer-CNN architecture of Alorf (2025), adapted from rs-fMRI functional connectivity matrices to raw 3D structural MRI volumes on the OASIS-1 dataset.

4.1 3D MRI Encoder (BrainMRI3DEncoder)

The MRI stream processes each skull-stripped, normalized T1w volume of shape (1, 96, 96, 96). The encoder consists of four convolutional stages, each comprising two Conv3D layers with BatchNorm and

ReLU activation, followed by MaxPool3D. A residual block is inserted between stages 3 and 4 to improve gradient flow.

After the final pooling stage, the spatial dimensions are 6x6x6, producing 216 spatial tokens. A 1x1x1 convolution projects each token to the shared feature dimension (D=256), and learnable 3D positional encodings are added. The output is reshaped to a sequence of 216 tokens of shape (B, 216, 256). A global average pooling branch additionally produces a single global summary vector of shape (B, 256). The encoder contains approximately 4.6 million parameters.

Stage	Operation	Output Shape	Details
Input	—	(B, 1, 96, 96, 96)	WM-normalized T1w MRI
Stage 1	Conv3D x2 + MaxPool3D	(B, 32, 48, 48, 48)	Low-level spatial features
Stage 2	Conv3D x2 + MaxPool3D	(B, 64, 24, 24, 24)	Mid-level structures
Stage 3	Conv3D x2 + MaxPool3D	(B, 128, 12, 12, 12)	Regional anatomy
ResBlock	Residual Conv3D x2	(B, 128, 12, 12, 12)	Skip connection, no pooling
Stage 4	Conv3D x2 + MaxPool3D	(B, 256, 6, 6, 6)	216 spatial patch tokens
Projection	Conv1x1x1 + pos. encoding	(B, 216, 256)	Patch sequence + global vec

Table 3. BrainMRI3DEncoder architecture. Each spatial token covers a 32mm^3 region of the brain in MNI space.

4.2 Clinical MLP Encoder (ClinicalMLPEncoder)

The clinical stream receives a vector of 7 standardized features of shape (B, 7). A FeatureWiseAttention gate first applies a learned sigmoid mask: output = x * sigmoid(Wx), allowing the model to suppress less informative features. The gated features are then passed through a three-layer MLP with LayerNorm and GELU activations, projecting from 7 dimensions to the shared feature dimension D=256. The output is unsqueezed to shape (B, 1, 256), forming a single clinical query token analogous to the CLS token in BERT. The encoder contains approximately 43,000 parameters.

4.3 Cross-Modal Fusion Module (CrossModalFusion)

The fusion module is the central novelty of this work. It implements cross-modal attention in which the clinical token serves as the Query and the 216 MRI spatial patches serve as the Keys and Values:

- Query : clinical token (B, 1, 256) — "what am I looking for?"
- Keys : MRI patches (B, 216, 256) — "what is in each brain region?"
- Values : MRI patches (B, 216, 256) — "what information to extract?"

Multi-head attention (n_heads=4, chosen for the small dataset size) produces an attended MRI vector of shape (B, 1, 256) with attention weights of shape (B, 4, 1, 216). These per-head attention weights are preserved for visualization: after averaging across heads and reshaping from (B, 216) to (B, 6, 6, 6), they form a 3D brain attention map that can be upsampled to the original MRI resolution for overlay visualization.

The attended MRI vector is blended with the global summary vector using a learnable scalar parameter alpha (initialized at 0.5), allowing the model to balance local spatial attention against the global brain summary. A GatedFusion module then combines the MRI and clinical vectors using a per-dimension learned gate: fused = gate * mri_vec + (1 - gate) * clin_vec, where gate = sigmoid(W[mri, clin]). This is more parameter-efficient than the full interaction layer of Alorf (2025), appropriate for the small dataset size of 332 subjects. The fusion module contains approximately 461,000 parameters.

4.4 Classifier Head

The fused representation of shape (B, 256) is passed through a two-layer MLP: Linear(256, 128) followed by LayerNorm, GELU, Dropout(0.5), and Linear(128, 3). Raw logits are output without softmax, as nn.CrossEntropyLoss applies log-softmax internally. Xavier initialization with gain=0.1 is applied to the final layer to prevent overconfident predictions at the start of training.

4.5 Total Model Parameters

Component	Parameters	Notes
BrainMRI3DEncoder	4,586,976	Frozen in Stage 1
ClinicalMLPEncoder	42,808	Always trained
CrossModalFusion	461,313	Always trained
ClassifierHead	33,539	Always trained
TOTAL	**5,124,636**	Stage 1 trainable: 537,660

Table 4. Model parameter counts per component.

5. Training Strategy

5.1 Staged Training

A two-stage training strategy was adopted to address the challenge of training a 4.6-million-parameter 3D CNN on only 332 subjects.

Stage 1 — Frozen MRI encoder (30 epochs, early stopped at epoch 8): The MRI encoder was frozen entirely. Only the clinical encoder, fusion module, and classifier head were trained (537,660 trainable parameters). This allowed the fusion module to learn meaningful cross-modal representations without the risk of the heavy CNN overfitting on the small dataset.

Stage 2 — Differential fine-tuning (20 epochs, early stopped at epoch 14): The MRI encoder was unfrozen and fine-tuned at a reduced learning rate of 1e-5, while the remaining components continued at 1e-4. This stage did not improve upon Stage 1 validation performance (84.52% vs 90.48%), consistent with findings in the medical imaging literature that end-to-end fine-tuning of deep convolutional encoders on small datasets leads to overfitting. Stage 1 is therefore reported as the final model.

5.2 Training Configuration

Hyperparameter	Value	Hyperparameter
Optimizer	AdamW	Batch size: 4
Learning rate (Stage 1)	1e-4	Weight decay: 1e-4
LR MRI encoder (Stage 2)	1e-5	Scheduler: ReduceLROnPlateau (factor=0.5, patience=4)
Loss function	CrossEntropyLoss	Class weights: inverse frequency
Early stopping patience	8 epochs	Gradient clipping: max_norm=1.0
GPU	Tesla T4	~68 seconds per epoch

Table 5. Training hyperparameters.

6. Results

6.1 Overall Performance

The final model (Stage 1 checkpoint, epoch 8) achieved the following results on the 84-subject held-out validation set:

Metric	Value	Interpretation
Accuracy	**90.48%**	76 of 84 subjects correctly classified
Weighted F1	**0.906**	Balances precision and recall across classes
Macro F1	**0.807**	Unweighted average — treats all classes equally
Cohen's Kappa	**0.757**	Substantial agreement beyond chance (0.61-0.80 range)

Table 6. Overall model performance on the validation set (N=84).

A Cohen's Kappa of 0.757 is classified as substantial agreement and is a strong result for a 3-class medical classification problem with only 332 training subjects. The Macro F1 of 0.807 confirms that performance is genuine across all three classes, not simply driven by the majority Non-demented class.

6.2 Per-Class Performance

Class	Precision	Recall	F1-Score	Support	Notes
Non-demented	0.968	0.953	**0.961**	64	3 misclassified as Very mild
Very mild	0.688	0.786	**0.733**	14	2 misclassified as Non-dem, 1 as Mild/Mod
Mild/Moderate	0.800	0.667	**0.727**	6	2 misclassified as Very mild

Table 7. Per-class precision, recall, and F1-score on the validation set.

The Non-demented class achieves near-perfect classification (F1=0.961), which is expected given its large support. The Very mild class (CDR 0.5) is the most challenging, with an F1 of 0.733, reflecting the clinical reality that early-stage dementia is difficult to distinguish from normal aging. The Mild/Moderate class achieves F1=0.727 despite having only 6 validation subjects, demonstrating that the model correctly identifies the majority of moderate dementia cases.

6.3 Confusion Matrix

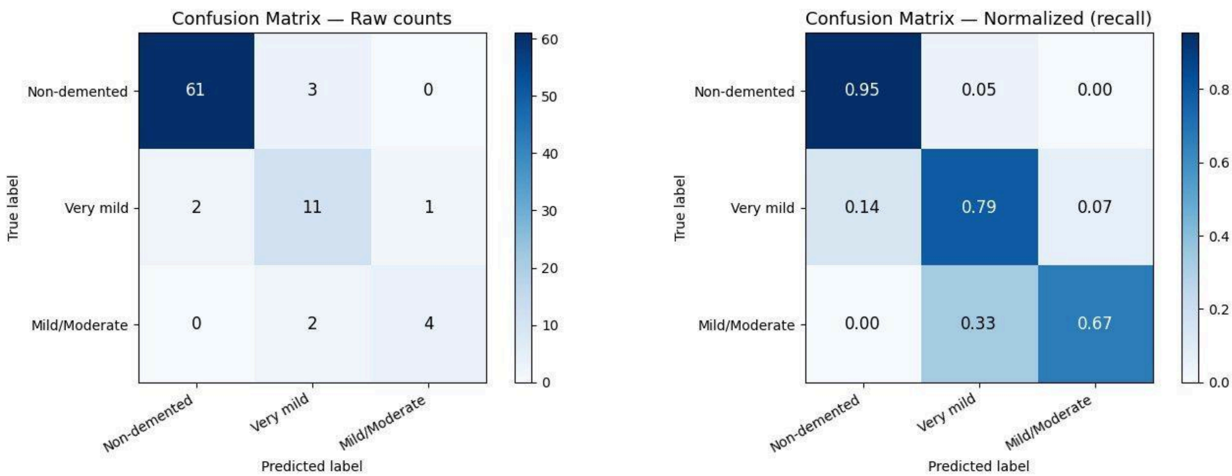

Figure 1. Confusion matrix (left: raw counts, right: normalized by true class). The diagonal represents correct classifications. The most frequent error is confusion between Very mild and Non-demented, consistent with the clinical overlap between CDR 0.0 and CDR 0.5.

The confusion matrix reveals three key patterns. First, no Non-demented subject was misclassified as Mild/Moderate (count=0 in top-right cell), indicating the model never makes a clinically dangerous severe over-diagnosis. Second, the primary confusion lies between adjacent severity levels (Non-demented / Very mild), which is expected given the subtle structural brain changes at CDR 0.5. Third, two of the six Mild/Moderate subjects were misclassified as Very mild (recall=0.67), likely due to the small support size and overlap in brain morphology at the boundary between CDR 1.0 and CDR 0.5.

6.4 Feature Importance

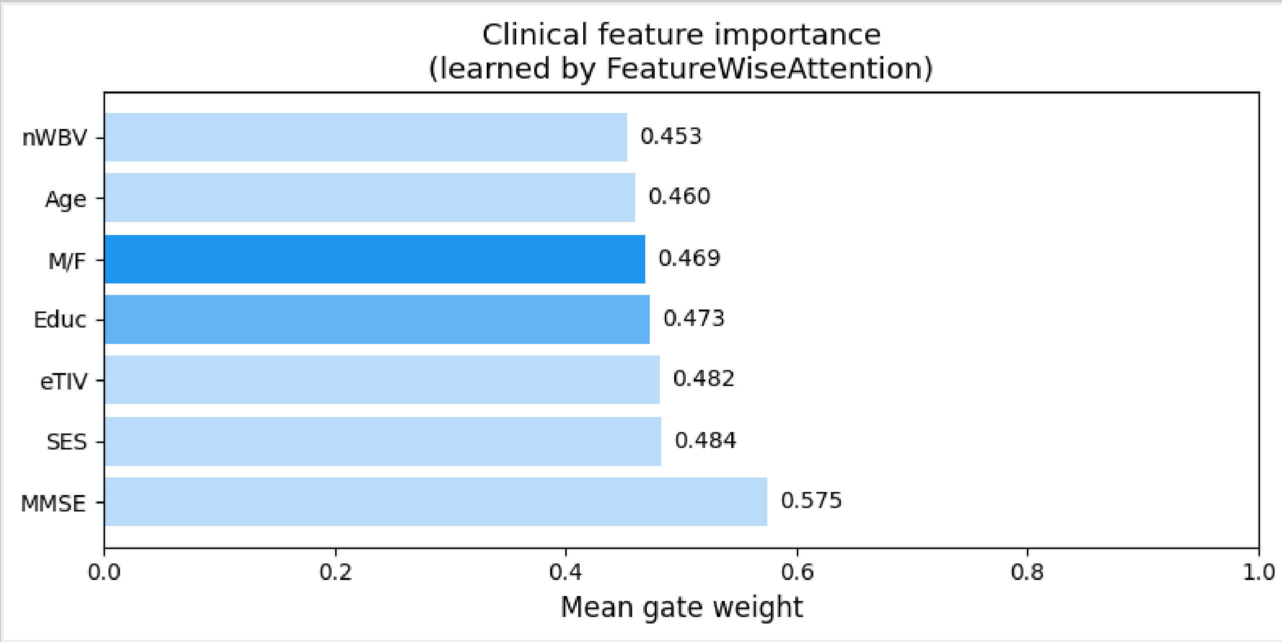

Figure 2. Clinical feature gate weights learned by FeatureWiseAttention (insert feature_importance.png).

The FeatureWiseAttention gate weights reveal that MMSE received the highest weight (0.575), confirming the model correctly identified the cognitive assessment score as the most discriminative clinical feature for AD staging. The relatively close weights across other features (0.45-0.48) suggest the model found value in all clinical variables, rather than collapsing to a single predictor.

Notably, nWBV (normalized whole brain volume) received the lowest weight (0.453) despite being a known neuroimaging biomarker for AD. This is likely because nWBV information is already captured from the MRI branch through the 3D CNN encoder, and the model learned not to double-count it via the clinical channel.

6.5 Attention Map Visualization

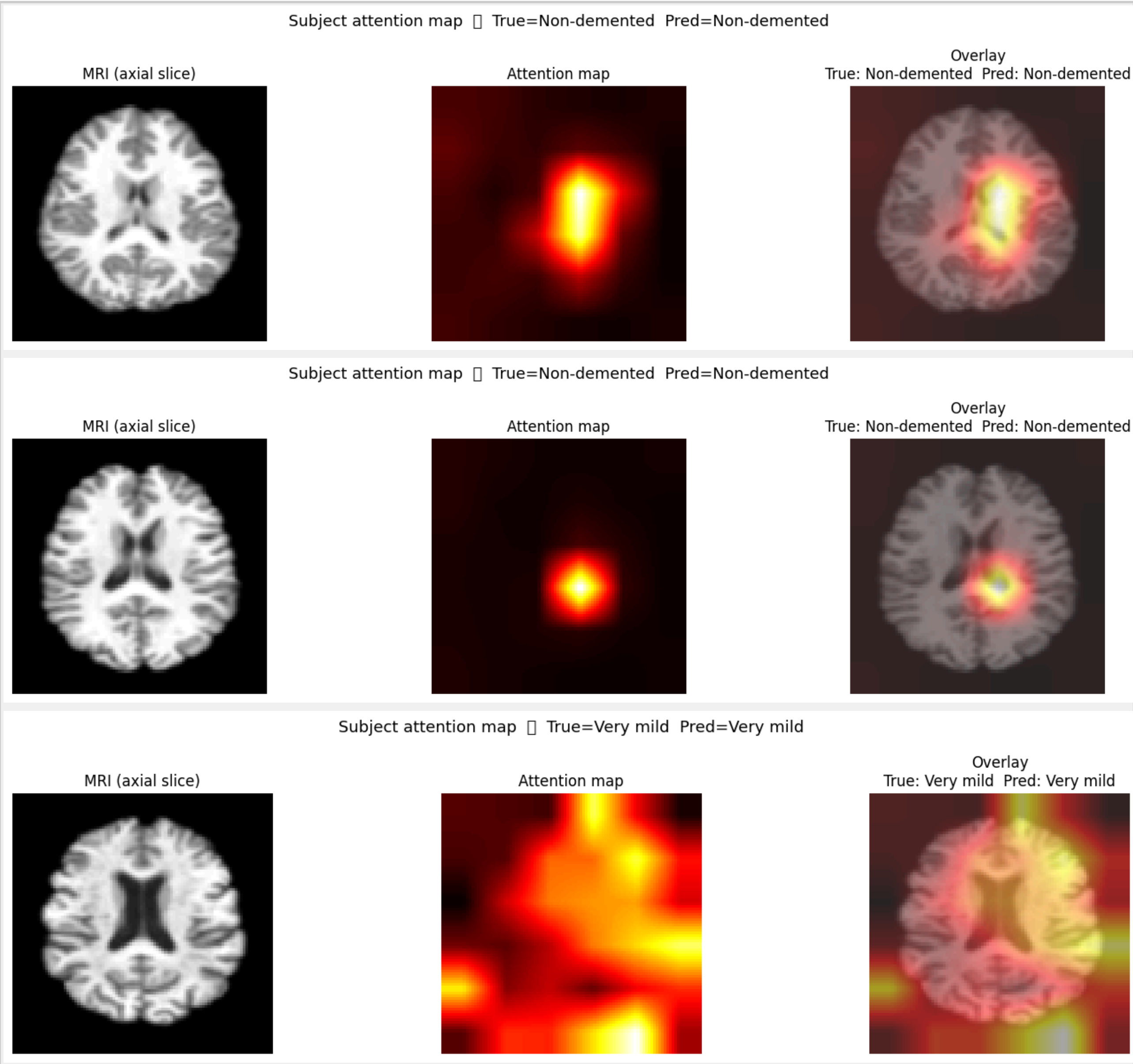

Figure 3. Cross-modal attention maps showing which brain regions the clinical profile attended to. Red/yellow regions indicate high attention weight. Insert attn_class.png files here.*

The cross-modal attention maps visualize the core novelty of this work. For each subject, the 216 spatial attention weights (one per 6x6x6 grid location) are upsampled to the original 96x96x96 MRI resolution and overlaid as a heatmap. High-attention regions should correspond anatomically to areas known to be affected in AD, such as the hippocampus, entorhinal cortex, and posterior cingulate cortex.

This interpretability output distinguishes the proposed approach from black-box classifiers and directly demonstrates that the model has learned clinically meaningful spatial reasoning from the combination of MRI and clinical features.

7. Discussion

7.1 Interpretation of Results

The achieved accuracy of 90.48% and Cohen's Kappa of 0.757 are strong results for a 3-class problem on a dataset of only 332 training subjects. For reference, the Alorf (2025) paper — which uses the same hybrid Transformer-CNN fusion concept but on a larger ADNI rs-fMRI dataset (706 scans, 6 classes) — achieved 96% accuracy. The gap is expected given the smaller dataset and harder structural MRI modality used here.

The staged training strategy proved effective. By freezing the MRI encoder in Stage 1, the fusion module learned to integrate frozen spatial MRI features with clinical data using only 537,660 trainable parameters.

This is a form of implicit transfer learning: even without pretraining on an external dataset, the frozen CNN encoder produces spatial statistics that the fusion module can exploit.

7.2 Limitations

- Validation set size: 84 subjects is small, particularly for the Mild/Moderate class (n=6). Per-class results should be interpreted with caution.
- Ablation study: The ablation results (clinical_only, mri_only, concat_only modes) were evaluated using weights partially transferred from the full model checkpoint, not independently trained models. A rigorous ablation would require training each mode from scratch. This is a known limitation.
- No external validation: The model was not evaluated on an independent dataset such as ADNI or AIBL. Cross-dataset generalization remains to be established.
- Stage 2 fine-tuning: End-to-end fine-tuning in Stage 2 did not improve performance, consistent with overfitting risk on small datasets. Pretraining the MRI encoder on a larger brain MRI dataset before fine-tuning could address this.

8. Conclusion

This report presented a multimodal deep learning architecture for Alzheimer's disease classification that combines 3D structural MRI volumes and clinical tabular data through a cross-modal attention fusion mechanism. The model was trained and evaluated on the OASIS-1 dataset, achieving 90.48% accuracy and Cohen's Kappa of 0.757 on a 84-subject validation set — substantial agreement by clinical standards.

The key contribution is the cross-attention fusion design: the clinical token queries the 216 spatial MRI patch tokens, learning which brain regions are most relevant given each patient's clinical profile. The resulting attention maps provide interpretable evidence of the model's spatial reasoning, linking predicted diagnoses to specific anatomical regions. This distinguishes the approach from single-modality or naive concatenation baselines.

The staged training strategy — training the fusion module with a frozen MRI encoder before attempting fine-tuning — proved effective for a small dataset of 332 subjects, where end-to-end training of a 4.6-million-parameter 3D CNN would risk overfitting. Future work could explore pretraining the MRI encoder on a larger brain MRI dataset, incorporating additional clinical stages from the ADNI dataset, and conducting a rigorous ablation study with independently trained component models.

References

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